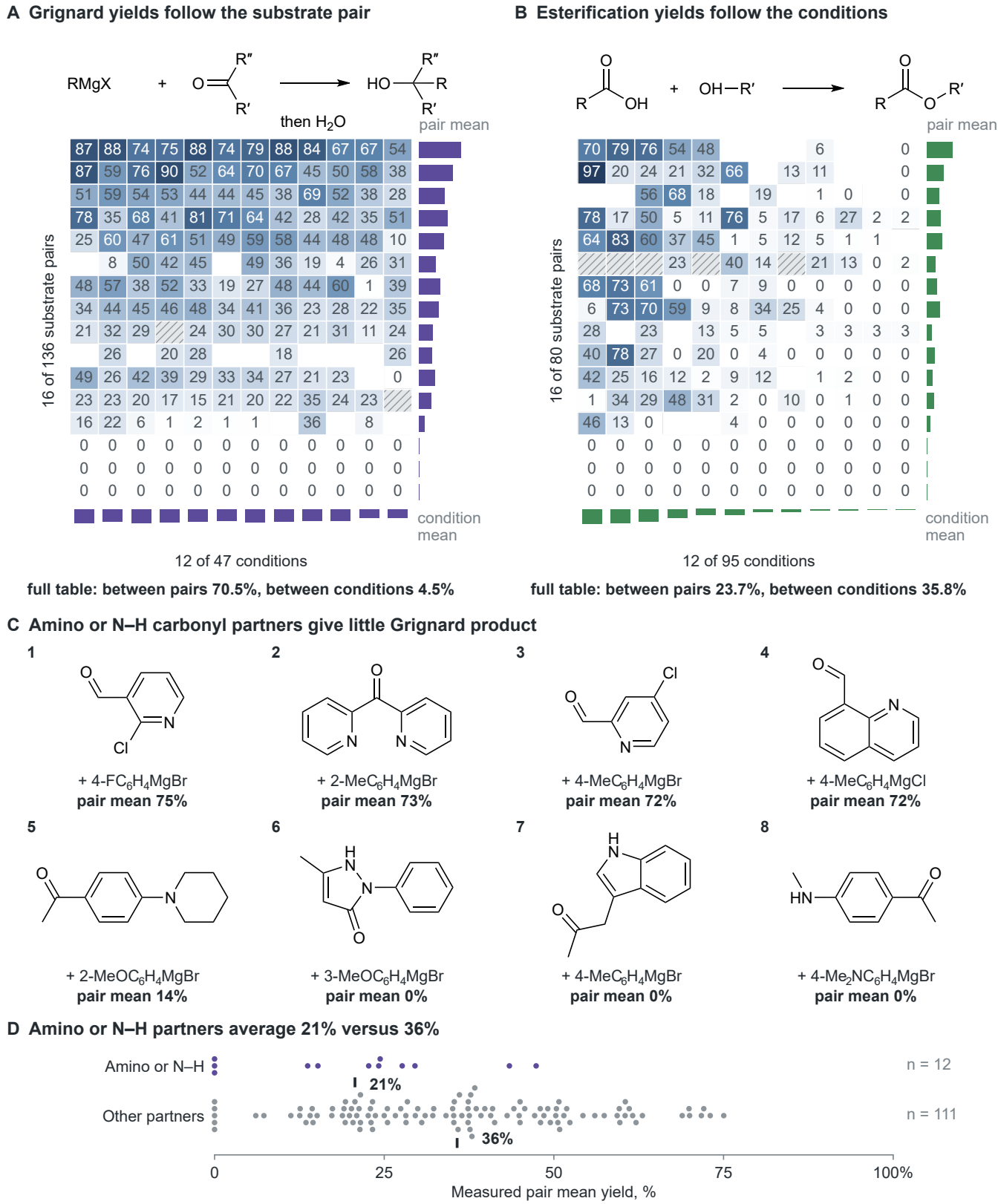

A Grignard yields follow the substrate pair
RMgX + O=C(R″)R′
then H2O
HO–C(R″)(R)R′
pair mean
16 of 136 substrate pairs
condition mean
12 of 47 conditions
full table: between pairs 70.5%, between conditions 4.5%
B Esterification yields follow the conditions
pair mean
16 of 80 substrate pairs
condition mean
12 of 95 conditions
full table: between pairs 23.7%, between conditions 35.8%
C Amino or N–H carbonyl partners give little Grignard product
1
+ 4-FC6H4MgBr
pair mean 75%
2
+ 2-MeC6H4MgBr
pair mean 73%
3
+ 4-MeC6H4MgBr
pair mean 72%
4
+ 4-MeC6H4MgCl
pair mean 72%
5
+ 2-MeOC6H4MgBr
pair mean 14%
6
+ 3-MeOC6H4MgBr
pair mean 0%
7
+ 4-MeC6H4MgBr
pair mean 0%
8
+ 4-Me2NC6H4MgBr
pair mean 0%
D Amino or N–H partners average 21% versus 36%
Amino or N–H
n = 12
21%
Other partners
n = 111
36%
0
25
50
75
100%
Measured pair mean yield, %